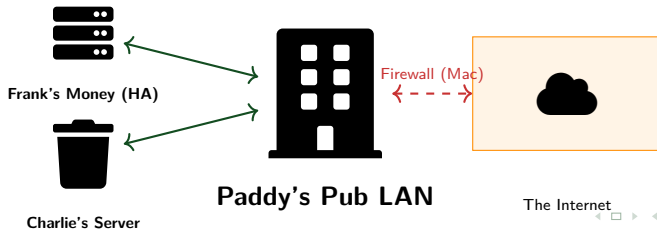


Module 7.0: Network Services & High Availability

"The Gang Configures the Network"

Brendan Shea, PhD

Rochester Community and Technical College



Review: The Gang Recaps Module 6

Previous Concepts

- **TCP/UDP:** Transport protocols. Reliable (TCP) vs Fast (UDP).
- **DHCP:** Dynamic Host Configuration Protocol. Assigns IPs.
- **DNS:** Domain Name System. Resolves names.

The New Problem

We have connectivity, but...

- Is the connection secure? (Can the McPoyles spy on us?)
- Is the time synced? (Are we opening on time?)
- Is the data backed up? (What if Dennis burns it down?)

Module 7 Objectives: The Scheme

1. Security & Time (Mac)

- **TLS**: Encryption/Authentication.
- **NTP**: Time Synchronization.

2. Web & Files (Dennis)

- **HTTP/HTTPS**: Web Services.
- **FTP/SMB**: File Transfer.

3. Email & Voice (Dee)

- **SMTP/IMAP**: Email Protocols.
- **VoIP**: Voice Services.

4. High Availability (Frank)

- **RAID/Clustering**: Redundancy.
- **DR**: Disaster Recovery.

"The Gang Secures the Network"

cue music...

Core Concepts

- **TLS**: Transport Layer Security. Encrypts traffic (e.g., HTTPS).
- **NTP**: Network Time Protocol. Syncs clocks for logs and auth.

TLS: Transport Layer Security

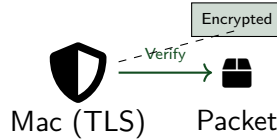
Definition

TLS is a cryptographic protocol designed to provide:

- 1 **Encryption:** Privacy (No eavesdropping).
- 2 **Integrity:** Data has not been changed.
- 3 **Authentication:** Verifying identity.

Analogy: Mac's "Ocular Patdown"

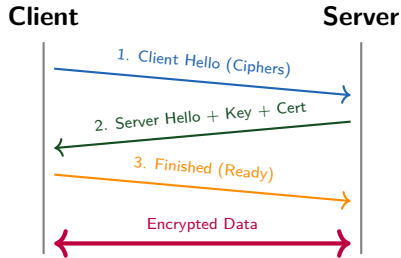
Mac assesses a threat (Authentication), ensures they aren't carrying weapons (Integrity), and clears them for entry.



Warning

SSL is insecure. Always use **TLS 1.2** or **1.3**.

TLS 1.3 Handshake: The Process



Technical Steps

- ① **Client Hello:** "I speak these ciphers."
- ② **Server Hello:** "Use this one. Here is my Cert."
- ③ **Key Exchange:** Generating session keys.
- ④ **Finished:** Secure tunnel up.

Analogy

The "Secret Handshake" before the gang discusses the scheme.

Certificates and Trust

Digital Certificates (X.509)

Binds an identity (Domain Name) to a Public Key.

- **CA (Certificate Authority):** Trusted 3rd party.
- **Chain of Trust:** Root CA -> Intermediate -> Server Cert.



Self-Signed Certificates

Signed by itself, not a trusted CA.

- **Result:** Browser warning ("Not Secure").
- **Use Case:** Internal testing only.

Analogy: Bird Law

A Certificate is a contract. You need a Judge (CA) to enforce it. Charlie's "Bird Law" (Self-Signed) is not recognized in court.

NTP: Network Time Protocol

Definition

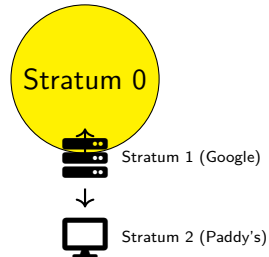
Protocol used to synchronize clocks.

- **Port:** UDP 123.
- **Why?:** Log correlation, Auth (Kerberos), Backups.

Stratum Levels

Hierarchy of distance from source.

- **Stratum 0:** Atomic Clock (The Source).
- **Stratum 1:** Directly connected to S0.
- **Stratum 16:** Unsynchronized (Desync).



Analogy: Charlie Work

Charlie ensures all clocks match so the bar opens exactly at 11 AM.

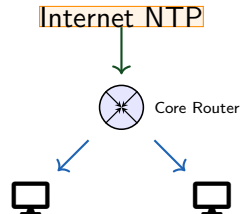
NTP Architecture: Hierarchy

Best Practice Setup

Don't have every PC query the internet.

- 1 **Router** queries Public NTP (pool.ntp.org).
- 2 **Switch** queries Router.
- 3 **PC/Server** queries Switch/Router.

This creates a single internal time source.



Precision Time Protocol (PTP)

When Milliseconds Aren't Good Enough

NTP is accurate to milliseconds (ms). **PTP (IEEE 1588)** is accurate to **microseconds (μs)** or even nanoseconds.

Use Cases

- **5G:** Cellular Sync.
- **Finance:** High-frequency trading.
- **Automation:** Robotics.

Analogy: The Nightman Cometh

In a musical, lights and audio must sync perfectly. Being off by 1ms ruins the show.



Case Study: Dennis & The Paddy's Time Heist

"The Gang's POS System Fails"

Paddy's POS system is rejecting credit cards with error: **"Certificate not yet valid."**

Facts:

- Current Date: June 1, 2025.
- System Date: **Jan 15, 1970** (Mac unplugged it).
- Cert Valid From: Jan 1, 2024.

"The computer thinks the internet hasn't been invented yet!" - Charlie

Review Questions

- 1 Why does the wrong date break TLS?
- 2 What protocol is missing?
- 3 How do we fix it permanently?

Case Study Solution: The Paddy's Time Heist

Solution: "The Gang Learns About Time"

- 1 **Validity:** The system thinks it is 1970. The certificate (from 2024) is "from the future" and thus invalid.
- 2 **Protocol: NTP** (Network Time Protocol).
- 3 **Fix:** Configure the POS to sync with `pool.ntp.org`.

The Fix

```
# chrony.conf  
server 0.pool.ntp.org iburst
```

Lesson

If time is broken, **Security is broken**. (Logs, Auth, and Certs all fail).

"The Gang Goes Online"

What we will cover

- **Web Browsing:** How Chrome talks to Servers (HTTP/HTTPS).
- **File Transfer:** Moving files across the internet (FTP).
- **File Sharing:** Accessing folders on the office network (SMB).
- **Storage:** Where the data actually lives (NAS/SAN/DB).

HTTP: Hypertext Transfer Protocol

What is it?

The language web browsers use to request webpages.

- **Port:** TCP 80.
- **Nature:** **Cleartext** (Not encrypted).

Analogy: Shouting in the Bar

If Charlie shouts "What is the Wi-Fi password?" across the bar, **everyone** hears it.

- **Client:** Charlie (Browser).
- **Server:** Dennis (Web Server).
- **Risk:** Frank (Hacker) is listening.

The Conversation (Request/Response)

Browser (GET):

GET /menu.html HTTP/1.1

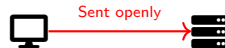
Host: www.paddyspub.com

Server (Response):

HTTP/1.1 200 OK

Content-Type: text/html

<html><body><h1>Wolf Cola</h1>...



HTTPS: HTTP Secure

What is it?

HTTP inside a secure, encrypted tunnel (TLS).

- **Port:** TCP 443.
- **Security:** Encrypted (Scrambled).
- **Indicator:** The  Lock icon in Chrome.

Analogy: The Back Office

Dennis takes the customer into the back office and locks the door to discuss "business."

- Frank can see them go in, but he **cannot hear** what they are saying.



HSTS

HTTP Strict Transport Security: A rule servers send to browsers saying "Never talk to me on Port 80 again. Only use Port 443."

HTTP Versions: Evolution of Speed

Why do we keep changing it?

The internet got heavier (images, videos). We needed faster ways to load pages.

HTTP/1.1 (1997)

"One at a time."

- Load image 1. Wait.
- Load image 2. Wait.
- **Slow.**

HTTP/2 (2015)

"Multiplexing."

- Load 10 images at once over one connection.
- Like carrying 10 beers on a tray.

HTTP/3 (2022)

"QUIC (UDP)."

- Abandon TCP!
- Uses UDP for raw speed.
- Fast setup (0-RTT).

FTP: File Transfer Protocol

What is it?

A protocol specifically for uploading/downloading files.

- **Port 21:** Control (Commands like "List files").
- **Port 20:** Data (The actual file content).

The Problem

FTP sends your **Username and Password** in plain text! *If Frank is sniffing the network, he gets your password immediately.*

Active vs. Passive

- **Active Mode:** The Server tries to connect back to the Client. *Firewalls block this ("Who is this server calling me?").*
- **Passive Mode (PASV):** The Client connects to the Server. *Firewalls like this (Client initiated it).*



Secure File Transfer: Don't use FTP!

SFTP (SSH File Transfer)

Recommended Standard.

- **Port:** TCP 22 (Same as SSH).
- **How:** It runs inside an SSH tunnel.
- **Pros:** Only 1 port to open on the firewall.

FTPS (FTP over SSL)

The Old Way.

- **Ports:** 989/990.
- **How:** Standard FTP wrapped in a certificate.
- **Cons:** Still uses multiple ports (Control/Data). Hard to firewall.

Exam Tip

If you see **SFTP**, think **SSH** (Port 22). If you see **FTPS**, think **SSL** (Certificates).

SMB: Server Message Block

What is it?

Windows File Sharing. This is what you use when you access a "Shared Folder" on the office network.

- **Port:** TCP 445.
- **Example:** \\PaddysServer\Schemes.
- **Mapping:** "The Z: Drive".

Security Warning: SMBv1

SMB Version 1 is extremely dangerous.

- It contains the **EternalBlue** vulnerability.
- This allowed the **WannaCry Ransomware** to spread worldwide in 2017.
- **Fix:** Disable SMBv1 on all Windows computers immediately.

Storage: NAS vs. SAN

Where does the data live?

When the server's hard drive is full, we buy dedicated storage.

NAS (Network Attached Storage)

"The Shared Box"

- **Type:** File Level.
- **Connection:** Ethernet (plug it into the switch).
- **Analogy:** A digital filing cabinet everyone can open.
- **Best for:** Office docs, backups.

SAN (Storage Area Network)

"The Virtual Disk"

- **Type:** Block Level.
- **Connection:** Fiber Channel (Expensive cables).
- **Analogy:** A hard drive attached via a very long cable. The server thinks it's a local disk.
- **Best for:** High-speed databases.

NAS Use Case: "The Evidence Locker"

Scenario

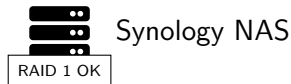
Dennis needs a place to store his... "videotapes."

- 1 He buys a Synology NAS.
- 2 He plugs it into the Paddy's Switch.
- 3 He creates a shared folder:
\\NAS\Evidence.
- 4 He sets permissions so only he can delete files.

RAID (Redundancy)

Dennis uses **RAID 1** (Mirroring) inside the NAS.

- If Drive A fails, Drive B still has the data.
- *"Two is one, and one is none."*



Databases: Frank's "Cooked Books"

SQL (Relational)

Structured Tables. Like an Excel sheet with strict rules.

- **Examples:** MySQL (port 3306), Microsoft SQL Server (port 1433).
- **Use:** Financial records, Inventory.
- **Frank's Query:** `SELECT * FROM Taxes WHERE Paid = 'False'`

NoSQL (Non-Relational)

Unstructured Data. Like a box of receipts.

- **Examples:** MongoDB, Redis.
- **Use:** Social media posts, caching, rapid data.
- **Benefit:** Very fast, flexible structure.

Security

Never expose your database port (e.g., 3306) to the internet. Hackers will brute-force the password in seconds. Keep it behind the firewall!

Case Study: Charlie's Rat Removal Website

"The Gang Gets Hacked"

Charlie launches `charliesratremoval.com` (Port 80). He collects credit card payments for his services. It works fine when he tests it locally. However:

- **Problem:** Customers see a red **"NOT SECURE"** warning.
- **Observation:** Frank intercepts credit card numbers in cleartext using Wireshark.

Discussion

- 1 Why was Frank able to steal the data?
- 2 What port/protocol must Charlie enable?
- 3 What does Charlie need to install?

Case Study Solution: Charlie's Rat Removal

Solution: "The Gang Encrypts"

- ❶ **Vulnerability:** HTTP (Port 80) sends data in **Cleartext**.
- ❷ **Fix:** Enable **HTTPS** on Port 443.
- ❸ **Requirement:** A **TLS Certificate** (e.g., Let's Encrypt).

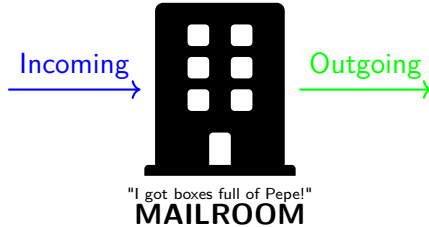
The Fix

```
$ sudo apt install certbot  
$ sudo certbot -apache
```

Result

Frank opens Wireshark again. He only sees encrypted garbage.

Section 7.3: Email & Voice (The Conspiracy)



Agents

- **MUA:** User Agent (Outlook, Thunderbird, Apple Mail).
- **MTA:** Transfer Agent (SMTP Server).
- **MDA:** Delivery Agent (IMAP/POP3 Server).

Protocols

- **SMTP:** Sending.
- **IMAP/POP3:** Receiving.

SMTP: Simple Mail Transfer Protocol

Sending Mail

SMTP is a **Push** protocol.

- **Port 25:** Server-to-Server. Shouldn't be used by clients, since it isn't secured.
- **Port 587:** Client-to-Server. Secured with TLS.
- **Port 465:** SMTP over SSL (Deprecated).

The Conversation

```
S: 220 paddys.com ESMTP
C: HELO google.com
S: 250 Hello
C: MAIL FROM: <dee@paddys.com>
C: RCPT TO: <waitress@coffee.com>
C: DATA
C: Subject: Bird
C: .
S: 250 Ok: queued
```

Spam Prevention

Use **SPF, DKIM, DMARC** to prove identity. These prevent spoofing by using DNS records and cryptographic signatures.

Receiving Mail: IMAP vs. POP3

IMAP (Sync)

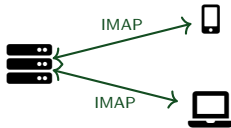
"The Cloud" (143/993)

- Stays on server.
- Synced on all devices.
- **Best for:** Dee, since she checks mail on phone & PC. (Multi-device).

POP3 (Download)

"Local Only" (110/995)

- Download & Delete.
- Local storage only.
- **Best for:** Frank, who only checks mail on one PC. (Single-device). He can free up server space, and archive locally.



VoIP: Voice Over IP (The Gang Starts a Call Center)

What is VoIP?

Sending voice as UDP packets.

- **Unified Comms:** Video/Voice/Chat.
- **Hard Phone:** Desk phone.
- **Soft Phone:** App (Teams).

Bandwidth

Codec	Speed
-------	-------

G.711	64 Kbps
-------	---------

Streaming 4K video kills

G.729	8 Kbps
-------	--------

VoIP (Jitter).



VoIP Protocols: SIP vs. RTP

SIP (Session Initiation)

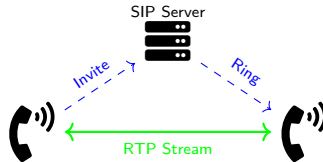
"The Setup" (5060/5061).

- Dialing/Ringing.
- "Hello?" / "Goodbye."

RTP (Real-time Transport)

"The Stream" (UDP).

- The actual audio data.
- Flows directly between phones.



VoIP Infrastructure: Power and VLANs

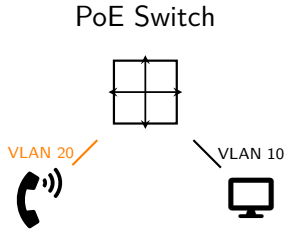
PoE (Power over Ethernet)

Sends power down the cable.

- **802.3af**: 15.4W.
- **802.3at**: 30W (Video).

Voice VLANs

- **VLAN 20**: Phones.
- **VLAN 10**: PCs.
- **Benefit**: Security + QoS.



Case Study: The Gang's Call Center

"Dee Sounds Like a Robot"

The Gang installs a cheap VoIP system to sell "Wolf Cola."

- **Setup:** IP Phones plugged into the same switch as Mac's gaming PC.
- **Issue:** When Mac downloads game updates, Dee's sales calls start stuttering and dropping.
- **Frank's Reaction:** "I'm not paying for this garbage! Fix it!"

Discussion Questions

- 1 What network phenomenon is causing the stuttering?
- 2 Why does Mac's download affect the phones?
- 3 What technology allows voice to skip the line?

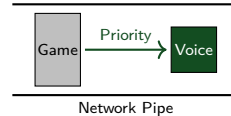
Case Study Solution: The Gang's Call Center

Solution: "The Gang Learns QoS"

- ❶ **Issue: Jitter** (Latency variation). Real-time voice cannot tolerate delays.
- ❷ **Cause: Congestion**. Mac is filling the bandwidth pipe, forcing voice packets to wait in the buffer.
- ❸ **Fix:** Voice and video traffic should be on a separate VLAN. The voice VLAN should implement **QoS (Quality of Service)** to ensure it has higher priority than best-effort traffic (like game downloads).

QoS Configuration

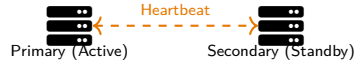
- **Tagging:** Mark voice packets with **DSCP EF (46)** (Expedited Forwarding).
- **Queuing:** Give EF packets strict priority over Best Effort (Game data).



High Availability: "The Show Must Go On"

Concepts

- **HA (High Availability)**: Redundancy to prevent downtime.
- **DR (Disaster Recovery)**: Plan for restoring after a catastrophe.



If Primary dies,
Secondary takes over!

Availability Metrics

Target	Downtime/Year
99%	3.65 days
99.9%	8.76 hours
99.999%	5 minutes

DR Metrics: RTO vs. RPO

How much does Frank lose?

When the server crashes, two clocks start ticking.

RPO (Recovery Point)

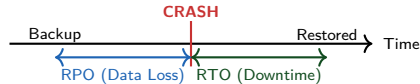
"Data Loss Tolerance"

- How far back do we go?
- If backup was 24h ago, we lost 24h of data.
- *Frank: "Zero!"*

RTO (Recovery Time)

"Downtime Tolerance"

- How long until we are back up?
- Includes repair and restore time.
- *Frank: "Now!"*



DR Sites: Paddy's Pub 2 (Electric Boogaloo)

Cold Site

"Empty Warehouse"

- Power/Net only.
- No hardware.
- **Slowest** / Cheapest.

Warm Site

"Storage Unit"

- Racks ready.
- Restore data needed.
- **Medium** cost.

Hot Site

"The Franchise"

- Exact Mirror.
- Instant failover.
- **Fastest** / Expensive.



Cold



Warm



Hot

Fault Tolerance: RAID

RAID (Redundant Array of Independent Disks) combines multiple physical drives into one logical unit for redundancy and/or performance.

RAID 0 (Striping)

"Charlie Special"

- Fast access.
- **Zero Redundancy.**
- 1 Drive fails = All data lost.

RAID 5 (Parity)

"The Gang Share"

- Data + Parity striped.
- Survives 1 failure.
- Efficient storage.

RAID 1 (Mirroring)

"Dennis System"

- Exact duplicate.
- 1 Drive fails = System stays up.
- 50% Capacity cost.



Parity protects data

FHRP: First Hop Redundancy Protocols

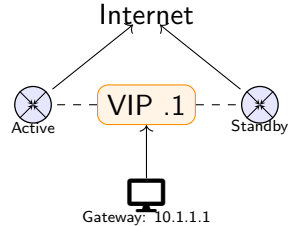
The Problem

If the main router (Gateway) dies, the bar loses internet.

The Solution: Virtual IP

Two routers share ONE IP address.

- **HSRP**: Cisco (Active/Standby).
- **VRRP**: Open (Master/Backup).
- **GLBP**: Load Balancing.



Case Study: Frank's "Foolproof" Plan

"The Gang Needs 5 Nines"

Frank demands **99.999%** availability for his gambling ring.

Current Setup:

- Single Consumer Router.
- Charlie manually backing up files to a USB stick "when he remembers."

Frank's Budget: \$50.

Discussion

- 1 Is "Five Nines" realistic?
- 2 Identify the **SPOF** (Single Point of Failure).
- 3 Suggest a realistic fix.

Solution: The "Good Enough" Plan

Solution: "Frank Compromises"

- ① **Reality Check:** 99.999% costs thousands (redundant ISPs, generators).
- ② **SPOF:** The router, the power, and Charlie.
- ③ **Realistic Fix:**
 - Buy a **UPS (Battery)** for power outages.
 - Automate backups to the **Cloud** (removes Charlie error).



UPS Battery

Module 7 Summary: The Gang Knows Networking

Services

- **TLS**: Secure the connection.
- **NTP**: Sync the time.
- **HTTPS**: Secure web.
- **SFTP**: Secure files.

High Availability

- **RAID**: Disk redundancy.
- **FHRP**: Router redundancy.
- **RTO/RPO**: Disaster planning.

"So anyway, I started pinging..." - Frank